

## Leverage and the Price-to-Earnings Multiple

This appendix demonstrates that the price-to-earnings (P/E) multiple of a levered company depends on its unlevered (all-equity) P/E, its cost of debt, and its debt-to-value ratio. When the unlevered P/E is less than  $1/k_d$  (where  $k_d$  equals the cost of debt), the P/E falls as leverage rises. Conversely, when the unlevered P/E is greater than  $1/k_d$ , the P/E rises with increased leverage.

In this proof, we assume the company faces no taxes and no distress costs. We do this to avoid modeling the complex relationship between capital structure and enterprise value. Instead, our goal is to show that there is a systematic relationship between the debt-to-value ratio and the P/E.

### STEP 1: DEFINING UNLEVERED P/E

To determine the relationship between P/E and leverage, start by defining the unlevered P/E ( $PE_u$ ). When a company is entirely financed with equity, its enterprise value equals its equity value, and its net operating profit after taxes (NOPAT) equals its net income:

$$PE_u = \frac{V_{ENT}}{NOPAT_{t+1}}$$

where  $V_{ENT}$  = enterprise value  
 $NOPAT_{t+1}$  = net operating profit after taxes in year  $t + 1$

This equation can be rearranged to solve for the enterprise value, which we will use in the next step:

$$V_{\text{ENT}} = \text{NOPAT}_{t+1} (\text{PE}_u) \quad (\text{D.1})$$

## STEP 2: LINKING NET INCOME TO NOPAT

For a company partially financed with debt, net income (NI) equals NOPAT less after-tax interest payments. Assuming the value of debt equals its face value, the company's interest expense will equal the cost of debt times the value of debt, which can be defined by multiplying enterprise value by the debt-to-value ratio:

$$\text{NI}_{t+1} = \text{NOPAT}_{t+1} - V_{\text{ENT}} \left( \frac{D}{V} \right) k_d$$

Substitute Equation D.1 for the enterprise value:

$$\text{NI}_{t+1} = \text{NOPAT}_{t+1} - \text{NOPAT}_{t+1} (\text{PE}_u) \left( \frac{D}{V} \right) k_d$$

Factor NOPAT into a single term:

$$\text{NI}_{t+1} = \text{NOPAT}_{t+1} \left[ 1 - \text{PE}_u \left( \frac{D}{V} \right) k_d \right] \quad (\text{D.2})$$

## STEP 3: DERIVING LEVERED P/E

At this point, we are ready to solve for the company's price-to-earnings ratio. Since P/E is based on equity values, first convert enterprise value to equity value. To do this, once again start with Equation D.1:

$$V_{\text{ENT}} = \text{NOPAT}_{t+1} (\text{PE}_u)$$

To convert enterprise value into equity value, multiply both sides by 1 minus the debt-to-value ratio:

$$V_{\text{ENT}} \left( 1 - \frac{D}{V_{\text{ENT}}} \right) = \text{NOPAT}_{t+1} (\text{PE}_u) \left( 1 - \frac{D}{V_{\text{ENT}}} \right)$$

Distribute  $V_{\text{ENT}}$  into the parentheses:

$$V_{\text{ENT}} - D = \text{NOPAT}_{t+1} (\text{PE}_u) \left( 1 - \frac{D}{V_{\text{ENT}}} \right)$$

Replace enterprise value ( $V_{ENT}$ ) minus debt ( $D$ ) with equity value ( $E$ ):

$$E = \text{NOPAT}_{t+1} (\text{PE}_u) \left( 1 - \frac{D}{V_{ENT}} \right)$$

Next, use Equation D.2 to eliminate  $\text{NOPAT}_{t+1}$ :

$$E = \frac{\text{NI}_{t+1} (\text{PE}_u) \left( 1 - \frac{D}{V} \right)}{1 - \text{PE}_u \left( \frac{D}{V} \right) k_d}$$

Divide both sides by net income to find the levered P/E:

$$\frac{E}{\text{NI}_{t+1}} = \frac{\text{PE}_u - \text{PE}_u \left( \frac{D}{V} \right)}{1 - \text{PE}_u \left( \frac{D}{V} \right) k_d}$$

At this point, we have a relationship between equity value and net income, which depends on the unlevered P/E, the debt-to-value ratio, and the cost of debt. Debt-to-value, however, is in both the numerator and the denominator, so it is difficult to distinguish how leverage affects the levered P/E. To eliminate the debt-to-value ratio in the numerator, use a few algebraic tricks. First, multiply both the numerator and denominator by  $k_d$ :

$$\frac{E}{\text{NI}_{t+1}} = \frac{\text{PE}_u (k_d) - \text{PE}_u \left( \frac{D}{V} \right) (k_d)}{k_d \left[ 1 - \text{PE}_u \left( \frac{D}{V} \right) (k_d) \right]}$$

Next, subtract and add 1 (a net difference of 0) in the numerator:

$$\frac{E}{\text{NI}_{t+1}} = \frac{[\text{PE}_u (k_d) - 1] + \left[ 1 - \text{PE}_u \left( \frac{D}{V} \right) (k_d) \right]}{k_d \left[ 1 - \text{PE}_u \left( \frac{D}{V} \right) (k_d) \right]}$$

After separating the numerator into two distinct terms, you can eliminate the components of the right-hand term by canceling them with the denominator. This allows you to remove debt-to-value from the numerator:

$$\frac{E}{\text{NI}_{t+1}} = \frac{\text{PE}_u (k_d) - 1}{k_d \left[ 1 - \text{PE}_u \left( \frac{D}{V} \right) (k_d) \right]} + \frac{1}{k_d}$$

To simplify the expression further, divide both the numerator and denominator of the complex fraction by  $k_d$ :

$$\frac{E}{NI_{t+1}} = \frac{1}{k_d} + \frac{PE_u - \frac{1}{k_d}}{1 - PE_u \left( \frac{D}{V} \right) (k_d)}$$

Finally, multiply the numerator and denominator of the second term by  $-1$ :

$$\frac{E}{NI_{t+1}} = \frac{1}{k_d} + \frac{\frac{1}{k_d} - PE_u}{\left( \frac{D}{V} \right) k_d (PE_u) - 1}$$

As this final equation shows, a company's P/E is a function of its unlevered P/E, its cost of debt, and its debt-to-value ratio. When the unlevered P/E equals the reciprocal of the cost of debt, the numerator of the second fraction equals zero, and leverage has no effect on the P/E. For companies with large unlevered P/Es, P/E systematically increases with leverage. Conversely, companies with small unlevered P/Es would exhibit a drop in P/E as leverage rises.